```
clc
clear all
close all
A=input("Enter the amplitude of the signal:");
fs=input("Enter the frequency:");
n=input("Enter the number of samples:");
f=fs/2;
t=linspace(0,4*pi,n);
msg=A*sin(t);
sgtitle("Nbit PCM Sarah John u2301189");
subplot(3,2,1)
plot(msg)
xlabel("Time");
ylabel("Amplitude");
title("Input signal");
```

Sampling of input signal

```
subplot(3,2,2)
stem(msg)
xlabel("Time");
ylabel("Amplitude");
title("Sampled signal");
```

Quantization of sampled signal

```
N=input("Enter the number of bits:");
L=2^N;
Vmax=max(msg);
Vmin=min(msg);
diff=Vmax-Vmin;
delta=diff/L;
part=Vmin:delta:Vmax;
code=Vmin-delta/2:delta:Vmax+delta/2;
[index,quants]=quantiz(msg,part,code);
subplot(3,2,3)
plot(quants);
xlabel("Time");
ylabel("Amplitude");
title("Quantized signal without correction");
```

Quantis and index correction

```
for i=1:length(msg)
    if quants(i)==Vmin-delta/2
        quants(i)=quants(i)+delta;
    end
    if index(i)>0
        index(i)=index(i)-1;
    end
```

```
end
subplot(3,2,4)
plot(quants)
xlabel("Time");
ylabel("Amplitude");
title("Quantized signal with correction");
```

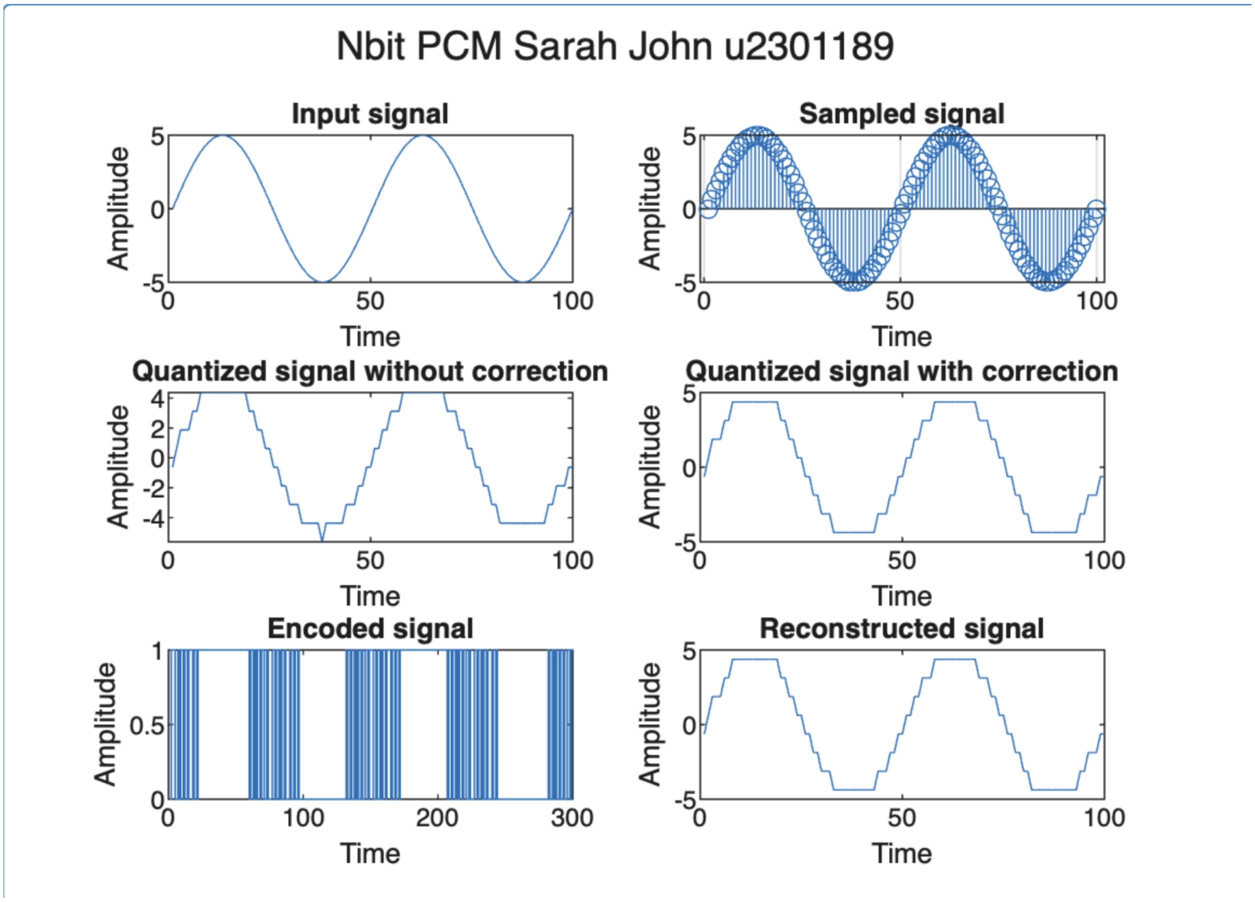
Encoding the quantized signal

```
y=int2bit(index,N);
B=reshape(y,1,[]);
subplot(3,2,5)
stairs(B)
xlabel("Time");
ylabel("Amplitude");
title("Encoded signal");
```

Decoding the signal to get original signal back

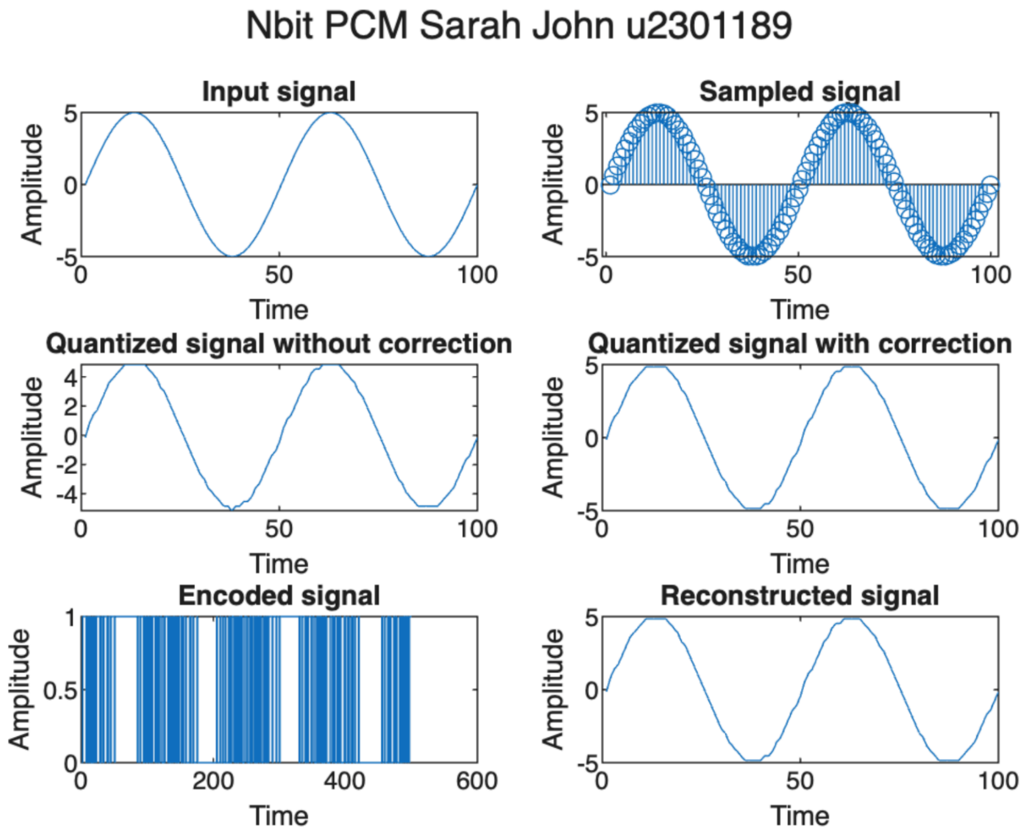
```
R=reshape(B,N,[]);
decode=bit2int(R,N);
for i=1:length(msg)
    reconst(i)=Vmin+delta/2+decode(i)*delta;
end
subplot(3,2,6)
plot(reconst);
xlabel("Time");
ylabel("Amplitude");
title("Reconstructed signal");
```

Case1 for N=3



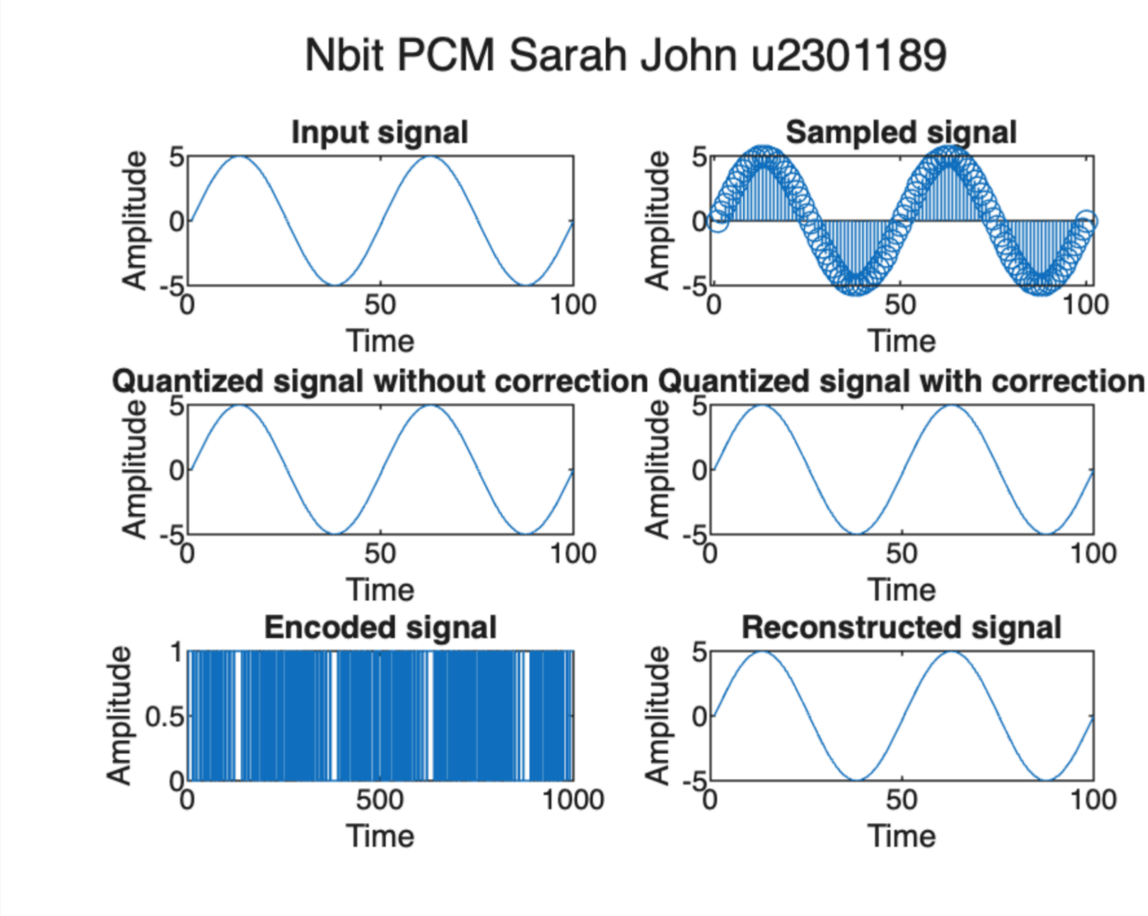

```
Enter the amplitude of the signal:5
Enter the frequency:1000
Enter the number of samples:100
Enter the number of bits:3
```

Case2 for N=5




```
Enter the amplitude of the signal:5
Enter the frequency:1000
Enter the number of samples:100
Enter the number of bits:5
```

Case3 for N=10




```
Enter the amplitude of the signal:5
Enter the frequency:1000
Enter the number of samples:100
Enter the number of bits:10
```